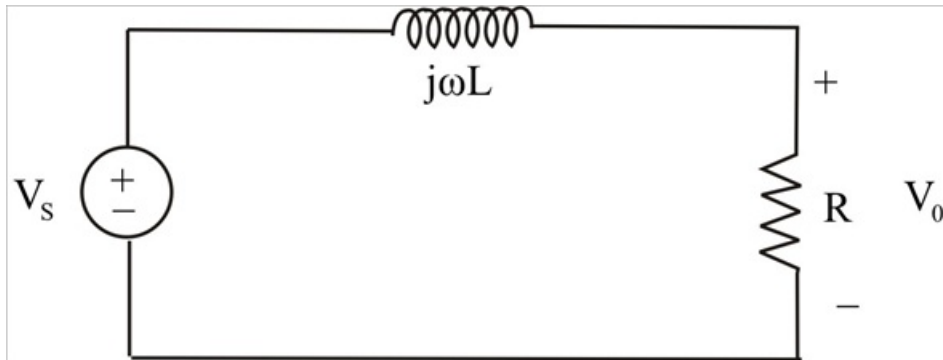


Problem

Design an RL low-pass filter that uses a 40-mH coil and has a cutoff frequency of 5 kHz.

Step-by-step solution

Step 1 of 5



Step 2 of 5

The transfer function of the circuit is.

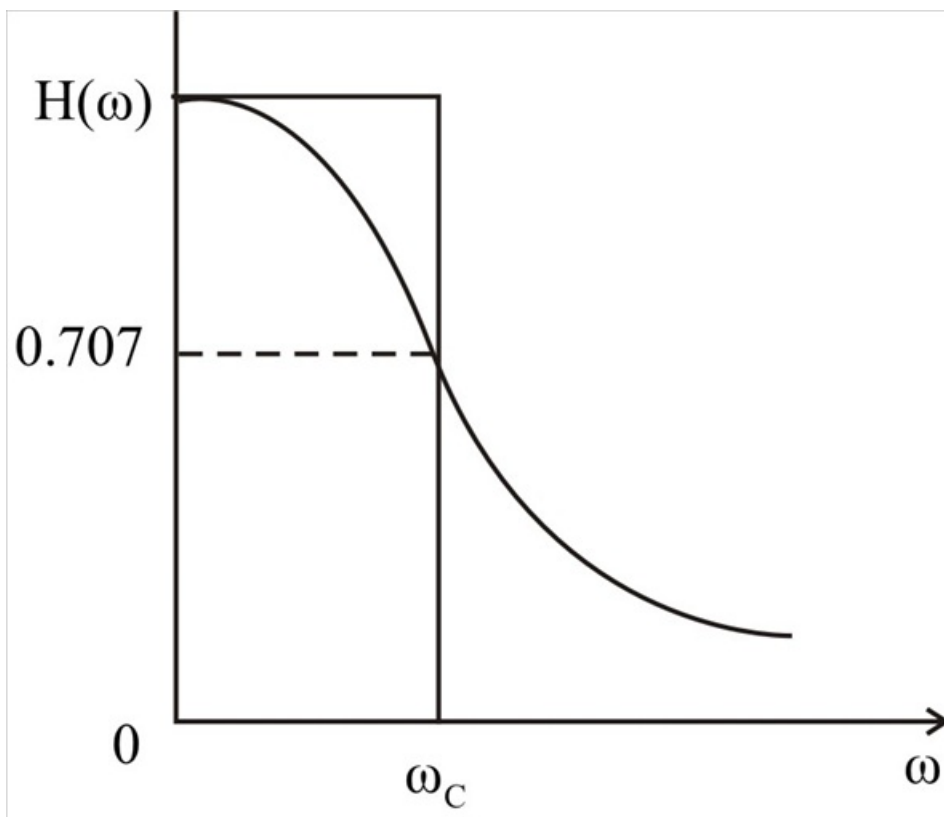
$$H(\omega) = \frac{V_o}{V_s} = \frac{\text{Out put voltage}}{\text{Input voltage}}$$

$$H(\omega) = \frac{R}{R + j\omega L}$$

$$H(\omega) = \frac{1}{1 + \frac{j\omega L}{R}}$$

$$H(0) = 1, \text{ \& } H(\infty) = 0$$

Step 3 of 5



Step 4 of 5

Magnitude of transfer function is,

$$|H(\omega)| = \frac{1}{\sqrt{1 + \frac{\omega^2 L^2}{R^2}}}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$

Step 5 of 5

Where,

$$\omega_c = \frac{R}{L}$$

$$2\pi f_c = \frac{R}{L}$$

Given that,

$$f_c = 5 \text{ kHz and}$$

$$L = 40 \text{ -mH}$$

Hence,

$$2\pi(5 \times 10^3) = \frac{R}{40 \times 10^{-3}}$$

$$R = 2\pi(5 \times 10^3) \times 40 \times 10^{-3}$$

$$\therefore \boxed{R = 1.256 \text{ k}\Omega}$$